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**KITCHEN RENOVATION PROJECT**

**OWNER: US CONSUL GENERAL OF CASABLANCA**

**DATE: 2 FEB 2026 FOR BID**

**SPECIFICATIONS**

- **PROTECTION OF EXISTING CONDITIONS**
- **ELECTRICAL**
- **PLUMBING**

## **PROTECTION OF EXISTING CONDITIONS**

### **PART 1 – GENERAL**

#### **1.1 Summary**

1. Provide temporary protection for character-defining architectural features and collections objects located outside the immediate construction zone but potentially impacted by contractor circulation, debris hauling, vibrations, and dust.
2. Limit vibrations, using hand tools where necessary, to minimize potential disturbance to sensitive, brittle and irreplaceable architectural finishes throughout the residence.

#### **1.2 References**

1. Property is a cultural heritage resource owned and operated by the USG, and as such all work shall be in compliance with the Secretary of Interior Standards for Historic Preservation.

#### **1.3 Submittals**

1. Protection Plan: Drawings and written description of proposed protection measures.
2. Product Data: Materials used for barriers, coverings, vibration monitoring devices.
3. Condition Documentation: Pre-construction photographs and written survey of features to be protected.

#### **1.4 Quality Assurance**

1. Contractor shall employ personnel experienced in protection of historic and sensitive materials.
2. Coordinate with Owner's representative and conservator for collections objects.

### **PART 2 – PRODUCTS**

#### **2.1 Protection Materials**

1. Dust Barriers: Flame-retardant, vapor-permeable (Tyvek) or offset (not taped) polyethylene sheeting, minimum 6 mil (0.15mm) thickness. Allow airflow to prevent condensation.
2. Vibration Monitoring: Portable seismographs or accelerometers, calibrated.
3. Surface Protection: Acid-free paper, padded blankets, plywood sheathing for impact-prone areas.
4. Floor Protection: Heavy-duty reinforced paper or composite board, slip-resistant.

## 2.2 Fasteners and Supports

1. Non-staining, non-invasive fasteners; avoid adhesives on historic surfaces.
2. Temporary framing: lightweight metal or chemically inert non-treated plywood or wood stud assemblies, removable without damage.
3. Miscellaneous supports, spacers, joints, and infill: chemically inert rigid load-bearing insulation

## PART 3 – EXECUTION

### 3.1 Preparation

1. Survey and document existing conditions prior to construction: **Please coordinate with OBO Office of Cultural Heritage for listing of heritage objects and requirements for protection, assessment and cataloguing.**
2. Identify circulation paths and debris hauling routes; install barriers accordingly.
3. Contractor shall limit construction vibrations to **≤0.14 in/sec (3.6 mm/s PPV)**, in accordance with **FTA Transit Noise and Vibration Impact Assessment Manual (2018)** Table 7.5 **Construction Vibration Damage Criteria**. For extremely fragile historic features, limit construction vibrations to **≤0.08 in/sec (2 mm/s PPV)**.

### 3.2 Installation

1. Erect dust barriers to isolate construction zones.
2. Cover sensitive surfaces with protective sheathing.
3. Install vibration monitoring devices adjacent to fragile features.

### **3.3 Maintenance**

1. Inspect protections daily; repair or replace damaged barriers immediately.
2. Maintain cleanliness; remove dust accumulations from protected areas.

### **3.4 Removal**

1. Remove protections only after completion of construction activities.
2. Restore protected features to pre-construction condition.

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## **DETAILED TECHNICAL SPECIFICATIONS (DTS)**

## **I. LOT: ELECTRICITY**

### **▪ Distribution Cables**

Low-voltage distribution from the main distribution panel (TGBT) to the sub-panels will be done using U1000 RO2V armored cables for external lighting ducts, following approved electrical schematics. These cables will be copper-core, with terminations crimped or screwed into cadmium-plated bolts.

- U1000 RO2V cables will be laid on cable trays, in a single layer, within conduits, ducts, voids in false ceilings, or embedded in conduits as necessary.
- A 2-meter slack will be left at designated locations specified by the related lot.
- Garden lighting cables (armored type) will be laid in double-wall corrugated pipes (not included in this price) from buried junction boxes.

The earth conductor will be identified by green-yellow double coloring.  
All circuits will be labeled with metal tags fixed near the cable terminations.

Before commissioning, all cables will undergo insulation testing without exception.

Without limitation, the contractor shall provide, install, and connect all low-voltage cables, including accessories, testing, fire-seal penetrations through fire-rated walls, and any additional requirements for a complete and fully operational primary distribution network.

The work will be measured per linear meter by cable type, including cables, connections, conduits, markings, and all supplies and labor. No additional cost will be claimed for omissions of any component or accessory essential for the proper operation of the system.

#### Linear Meter Prices

- **Price No. 1:** U1000 - RO2V – 3G6 mm<sup>2</sup> (PRICE PER LINEAR METER)
- **Price No. 2:** U1000 - RO2V – 3G2.5 mm<sup>2</sup> (PRICE PER LINEAR METER)
- **Price No. 3:** U1000 - RO2V – 3G1.5 mm<sup>2</sup> (PRICE PER LINEAR METER)

### **▪ Electrical Power Supplies**

The electrical power supply for all technical equipment, including electricity, elevators, low-voltage systems, plumbing, air conditioning, medical gases, kitchens, laundry, morgues, etc., will comply with the technical description below and the single-line schematics attached to this document. These two documents are complementary.

Electrical supplies will be provided up to the designated points and interfaces specified by the concerned lots. Locations on the plans are indicative and will be confirmed during the execution phase in collaboration with the relevant contractors. Cable sections and power ratings will be calculated by the contractor and validated by the control office and the design office.

Prices of electrical power supplies for technical equipment will include all necessary materials and labor for supply, cabling, tubing, protection, fastenings, seals, connections, and installation

according to applicable norms and best practices. No additional costs will be granted to ensure conformity with regulations and requirements.

**- Price No. 1: Split System Power Supply**

Includes the supply and installation of an appropriately sized U1000 RO2V cable per the attached single-line schematics. This includes waterproof junction boxes (Plexo type), electrical terminal blocks, and all accessories required for cable installation.

Cables will be installed under cable trays, embedded in ICD conduits, or through IRO (PVC) tubing of appropriate diameters, from the respective protection electrical panel to the connection box.

The completed work, including supply, installation, connection, and all accessories, will be paid as a complete unit.

**- Price No. 2: Intercom Power Supply**

Includes the supply, installation, and connection of U1000 RO2V or H07-VU cables of appropriate section per the attached single-line schematics. This includes waterproof junction boxes (Plexo type), terminal blocks, and all necessary protections and accessories.

Cables will be installed under cable trays, embedded in ICD conduits, or through IRO (PVC) tubing of appropriate diameters, from the respective protection electrical panel to the connection box.

The completed work, including supply, installation, and connection, will be paid as a complete unit.

▪ **Lighting Fixtures**

**Lighting Points**

The entire lighting and low-voltage distribution within the building will be installed using embedded ICD (ICTL) conduits within masonry or forms, or ICT (ICTA) conduits for non-embedded areas.

Conduit selection will comply with room specifications and regulations, with a minimum size of No. 13.

Wiring will be carried out using HO7V-U (U500V) conductors in conduits or U1000 RO2V cables on cable trays or raceways. The minimum conductor size for lighting will be 1.5 mm<sup>2</sup>, and 2.5 mm<sup>2</sup> or 4 mm<sup>2</sup> for low-voltage outlets.

**Embedded Low-Voltage Apparatus Includes:**

- Simple switches (SA)
- Two-way switches (VV)
- Push-buttons (BP)
- Electrical outlets
- Wire exits for 3x2.5 electrical connections
- Telephone and data outlets (CAT 6A)
- Junction and connection boxes

The embedded apparatus will be of the Mosaic Legrand or Unica Basic Schneider Electric brands or equivalent. The model, finish, and color, as well as type (screw-in or snap-in), will be selected by the architect.

**Waterproof Apparatus:**

Similar to the embedded apparatus but waterproof, it will be of brands such as Legrand Plexo, Schneider Electric Mureva, or Simon Aqua.

### **Industrial Apparatus:**

To be installed in areas specified by the design office, it will be of standardized industrial type such as the Legrand Hypra series.

The contractor must provide samples of all materials for approval before installation. Accepted samples will remain on-site as reference materials. All luminous points without installed fixtures will be terminated with a box, cover, wire exit, and a compact fluorescent lamp for initial use.

### **Specific Items:**

- Adjacent power and data outlets in office or hallway clusters will be grouped into modular faceplates, included in the outlet price.
- All cutting, trenching, wall or masonry drilling, and repairs to affected elements are included in the price.

### **Lighting Prices**

#### **- Price No. 3: Single-Lighting Point**

This price rewards:

- The line from the ICD n°13 or ICT n°13 duct distribution panel comprising a 3 U1000RO2V G1.5 mm<sup>2</sup> cable laid in the building voids to the light centre, stopped on a recessed reservation pot.
- The line from the fireplace to the ICD duct switch n°13 or ICT n°13 comprising 2 H07-VU 1x1.5mm<sup>2</sup> conductors or U1000RO2V 2x1.5mm<sup>2</sup> cables
- A switch reservation pot in the masonry.
- Galvanized iron wire in the ducts for pulling the conductors.
- The 10A single-ignition switch type recessed from the MOSAIC Legrand series, UNICA BASIC from SCHNEIDER ELECTRIC, the frame color is at the architect's choice.
- The light point with socket and wire outlet.
- The installation, connection and fixing of all equipment, including accessories, in accordance with the rules of the art and the plans.
- Tube inlet sleeves, wiring, grooves, ICD or ICT conduits, wiring, spackling, terminals, connections, fasteners and tests.

Work paid for individually including installation, mounting accessories, frame, connection, installation accessories, all supplies and conditions. No added value will be requested in the event of an omission in the description of a component or accessory necessary for the proper functioning of the structure.

#### **- Price No. 4: Two-Way Switch Lighting Point**

This price rewards:

- The line from the ICD n°13 or ICT n°13 duct distribution panel comprising a 3 U1000RO2V G1.5 mm<sup>2</sup> cable laid in the building voids to the light centre, stopped on a recessed reservation pot.
- The line from the fireplace to the ICD n°13 or ICT n°13 duct switch comprising 2 to 4 H07-VU 1x1.5mm<sup>2</sup> conductors or U1000RO2V 2x1.5 mm<sup>2</sup> or 4x1.5 mm<sup>2</sup> cables.
- The shuttle line from the first reciprocating switch to the second switch in ICD n°13 or ICT n°13 duct comprising 2 H07-VU 1x1.5mm<sup>2</sup> conductors or in cables U1000RO2V 2x1.5mm<sup>2</sup>.
- Two switch reservation pots in the masonry.

- Galvanized iron wire in the ducts for pulling the conductors.
- The 10A recessed type recessed switches of the MOSAIC Legrand series, UNICA BASIC of SCHNEIDER ELECTRIC or equivalent, the frame color is at the architect's choice.
- The light point with socket and wire outlet.
- The installation, connection and fixing of all equipment, including accessories, in accordance with the rules of the art and the plans.
- Tube inlet sleeves, wiring, grooves, ICD or ICT conduits, wiring, spackling, terminals, connections, fasteners and tests.

Work paid for individually including installation, mounting accessories, frame, connection, installation accessories, all supplies and conditions.

No added value will be requested in the event of an omission in the description of a component or accessory necessary for the proper functioning of the structure.

PRICE PAID PER UNIT.

**- Price No. 5: Additional Lighting Point**

This price rewards:

- The line from the first fireplace to the second furnace in ICD n° 13 or ICT n°13 duct comprising 3 H07-VU 1x1.5mm<sup>2</sup> conductors or 3x1.5 mm<sup>2</sup> U1000RO2V cables placed on in the construction voids, stopped on a recessed reservation pot.
- The galvanized iron wire in the sheath for pulling the conductors.
- The installation, connection and fixing of all equipment, including accessories, in accordance with the rules of the art and the plans.

PRICE PAID PER UNIT.

**- Price No. 6: Waterproof Power Outlet 2P+T 16A**

This price rewards:

- The line from the switchboard to the first socket base, comprising a 3G2.5 mm<sup>2</sup> cable U1000RO2V laid on a cable tray, on a duct or in the ICD n°13 or ICT n°13 duct construction voids, stopped on a recessed reservation pot.
- The line between different bases belonging to the same circuit in ICD n° 13 or ICT n°13 duct or on a duct comprising 3 H07-VU 1x2.5mm<sup>2</sup> conductors or in 3G2.5 mm<sup>2</sup> U1000RO2V cables laid on in the construction voids, stopped on a recessed reservation pot.
- Pots of reservation of the socket in the masonry.
- The IP55 waterproof IK07 2P+T 16A socket outlet with flap from the PLEXO series from LEGRAND, Mureva from Schneider Electric, the Aqua series from Simon or equivalent.
- Galvanized iron wire in the ducts for pulling the conductors.
- The installation, connection and fixing of all equipment, including accessories, in accordance with the rules of the art and the plans.
- Tube inlet sleeves, wiring, grooves, ICD or ICT conduits, wiring, spackling, terminals, connections, fasteners and tests.

Work paid for individually including installation, mounting accessories, frame, connection, installation accessories, all supplies and conditions.

No added value will be requested in the event of an omission in the description of a component or accessory necessary for the proper functioning of the structure.

PRICE PAID PER UNIT.

▪ **Command Panels**

**-Price No. 7: Command Panel**

The lighting control circuits for corridors and halls will be installed using command panels of the Legrand Mosaic or Simon 82 series, accommodating 6 to 12 modules with push-buttons or illuminated switches. Panels will be flush-mounted in locations indicated on plans.

The plate will be installed recessed in the place indicated on plans on a 6 or 12 module recessed box.

Work paid for by the unit including installation, mounting accessories, support, plate, connection, installation accessories, all supplies and conditions.

No added value will be requested in the event of an omission in the description of a component or accessory necessary for the proper functioning of the structure.

PRICE PAID PER UNIT.

▪ **Decorative Lighting Fixtures**

N.B: It must comply with CE standards and the luminaires must have protection classes that comply with the standards in force according to their location.

This price pays for the entire supply and installation of the chandelier elements, including all supply requirements (bulbs, deflectors, grilles, etc.) and installation (drilling, sealing, connection, etc.). The chandelier will be recessed or exposed, at the choice of the project manager.

The chandelier can only be installed after approval by the project owner and the project manager.

All the luminaires will be from the brands Thorn, Fosnova, Troll, 3F Fillipi, ideallux, Regent, Rovasi, LedC4, Disano, trilux, Philips or similar.

All auxiliary equipment (lamps, electronic ballasts, capacitors, ignitors and others) will be of original brands PHILIPS, OSRAM or similar.

No added value will be requested in the event of an omission in the description of an element, a service or a quality characteristic necessary for the proper functioning of the work.

**-Price No. 8: Suspended Decorative Luminaire**

Supply and installation of a decorative pendant lamp Decorative pendant light fixture with several circular light elements of 3 to 4 circles. Direct-indirect light intensity distribution. For individual suspended mounting, the length of the suspension wires will be determined by the project manager, protection provided by a primary screen. Glare assessment (EN 12464-1) according to UGR a > 80. Luminaire body made of sheet steel, powder lacquered, white. The diameter of each circle between Ø 75 mm and Ø 1200 mm, height of the luminaire at the architect's choice.

Work paid for individually including installation, mounting accessories, support, false ceiling reservation, connection, installation accessories, all supplies and conditions.

No added value will be requested in the event of an omission in the description of a component or accessory necessary for the proper functioning of the structure.

PRICE PAID PER UNIT.

900067  
MARGHERA-Z



General Information

|                               |                |
|-------------------------------|----------------|
| Material number               | 900067         |
| Type                          | ceiling light  |
| Product segment               | indoor         |
| Theme                         | EGLO connect.2 |
| Energy efficiency class (EEI) | E              |

Dimensions

|                             |      |
|-----------------------------|------|
| Article Height (in mm/in)   | 110  |
| Article Diameter (in mm/in) | 600  |
| Net Weight (in kg)          | 4.17 |

Material & Colour

|                      |         |
|----------------------|---------|
| Enclosure Material   | steel   |
| Enclosure Colour     | black   |
| Glass/Shade Material | plastic |
| Glass/Shade Colour   | white   |

Functionality

|             |                                              |
|-------------|----------------------------------------------|
| Switch Type | without switch                               |
| Function    | control via APP + remote control<br>CCT, RGB |
| Battery     | No                                           |

-Price No. 9: Round LED Spotlights

This price rewards the supply and installation of a light fixture: (See technical sheet below)

Work paid for individually including installation, mounting accessories, support, false ceiling reservation, connection, installation accessories, all supplies and conditions.

No added value will be requested in the event of an omission in the description of a component or accessory necessary for the proper functioning of the structure.

PRICE PAID PER UNIT.

99215  
FUEVA 5



General Information

|                               |                             |
|-------------------------------|-----------------------------|
| Material number               | 99215                       |
| Type                          | surface-mounted light       |
| Product segment               | indoor                      |
| Theme                         | sonstige / nicht zuordenbar |
| Energy efficiency class (EEI) | E                           |

Dimensions

|                             |      |
|-----------------------------|------|
| Article Height (in mm/in)   | 28   |
| Article Diameter (in mm/in) | 160  |
| Net Weight (in kg)          | 0.35 |

Material & Colour

|                      |         |
|----------------------|---------|
| Enclosure Material   | steel   |
| Enclosure Colour     | white   |
| Glass/Shade Material | plastic |
| Glass/Shade Colour   | white   |

Functionality

|             |                |
|-------------|----------------|
| Switch Type | without switch |
| Battery     | No             |

**-Price No. 10:** High-Temperature LED Strip (IP67)

This prize rewards:

- The line from the ICD n°13 or ICT n°13 duct distribution panel comprising a 3G U1000RO2V 1.5 mm<sup>2</sup> cable laid in the construction voids to the light strip, stopped on a recessed reservation pot.
- The line from the ribbon to the ICD n°13 or ICT n°13 duct switch comprising 4 H07-VU 1x1.5mm<sup>2</sup> conductors or 4x1.5mm<sup>2</sup> U1000RO2V cables.
- A switch reservation pot in the masonry.
- Galvanized iron wire in the ducts for pulling the conductors.
- The 10A single-ignition switch type recessed from the MOSAIC Legrand series, UNICA BASIC from SCHNEIDER ELECTRIC or equivalent, the frame color is at the architect's choice.
- The light point with socket and wire outlet.
- The installation, connection and fixing of all equipment, including accessories, in accordance with the rules of the art and the plans.
- Tube inlet sleeves, wiring, grooves, ICD or ICT conduits, wiring, spackling, terminals, connections, fasteners and tests.
- IP67 high-temperature tape of the architect's choice.

Work paid for individually including installation, mounting accessories, frame, connection, installation accessories, all supplies and conditions.

No added value will be requested in the event of an omission in the description of a component or accessory necessary for the proper functioning of the structure.

PRICE PAID PER UNIT.

**-Price No. 11:** Ceiling Light Fixture

This price rewards the supply and installation of a light fixture: (See technical sheet below)

Work paid for individually including installation, mounting accessories, support, false ceiling reservation, connection, installation accessories, all supplies and conditions.

No added value will be requested in the event of an omission in the description of a component or accessory necessary for the proper functioning of the structure.

PRICE PAID PER UNIT.

900275  
JABAGA



Informations générales

|                     |                        |
|---------------------|------------------------|
| Reference EGLO      | 900275                 |
| Type                | Applique               |
| Ségment de produits | Luminaire extérieur    |
| Sujet               | Autre / non associable |

Dimensions

|                                |                 |
|--------------------------------|-----------------|
| Hauteur de l'article (en mm)   | 80              |
| Longueur de l'article (en mm)  | 70              |
| Poids net (en kg)              | 0.14 kilogramme |
| Amplitude de l'article (en mm) | 90              |

Matériau et couleur

|                                   |             |
|-----------------------------------|-------------|
| Matériau principale               | plastique   |
| Couleur principale                | noir        |
| Matériau du verre/de l'abat-jour  | plastique   |
| Couleur du verre / de l'abat-jour | transparent |

Fonctionnalité

|                     |                   |
|---------------------|-------------------|
| Type d'interrupteur | sans interrupteur |
| Pile                | Aucun             |

**-Price No. 12: Round LED Spotlights**

This price rewards the supply and installation of a light fixture: (See technical sheet below)

Work paid for individually including installation, mounting accessories, support, false ceiling reservation, connection, installation accessories, all supplies and conditions.

No added value will be requested in the event of an omission in the description of a component or accessory necessary for the proper functioning of the structure.

PRICE PAID PER UNIT.

**900067**  
**MARGHERA-Z**



#### General Information

|                               |                |
|-------------------------------|----------------|
| Material number               | 900067         |
| Type                          | ceiling light  |
| Product segment               | indoor         |
| Theme                         | EGLO connect.2 |
| Energy efficiency class (EEI) | E              |

#### Dimensions

|                             |      |
|-----------------------------|------|
| Article Height (in mm/in)   | 110  |
| Article Diameter (in mm/in) | 600  |
| Net Weight (in kg)          | 4.17 |

#### Material & Colour

|                      |         |
|----------------------|---------|
| Enclosure Material   | steel   |
| Enclosure Colour     | black   |
| Glass/Shade Material | plastic |
| Glass/Shade Colour   | white   |

#### Functionality

|             |                                              |
|-------------|----------------------------------------------|
| Switch Type | without switch                               |
| Function    | control via APP + remote control<br>CCT, RGB |
| Battery     | No                                           |

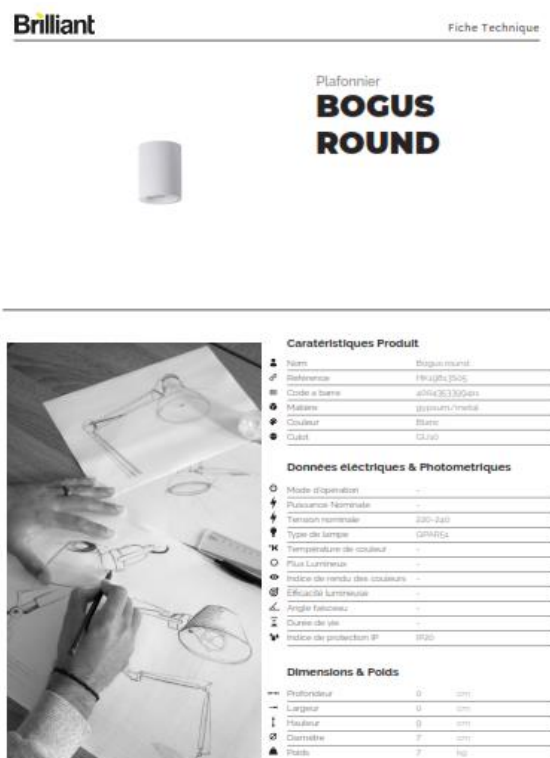
**-Price No. 13: LED Light Fixture Type 2**

This price rewards the supply and installation of a light fixture: (See technical sheet below)

Work paid for individually including installation, mounting accessories, support, false ceiling reservation, connection, installation accessories, all supplies and conditions.

No added value will be requested in the event of an omission in the description of a component or accessory necessary for the proper functioning of the structure.

PRICE PAID PER UNIT.



#### ▪ **Intercom Kit**

##### **-Price No. 14: Intercom Kit**

Includes the supply, installation, cabling, and commissioning of an intercom communication kit for easy communication between the kitchen and clients. The kit will feature:

- LCD screen
- Minimum protection rating: IP64 IK08
- PoE or external 24V power supply
- Numeric keypad and function keys
- IP communication
- HD voice (7 kHz)
- 1.5 W - 75 dB audio output at 1 meter

Intercoms will be installed according to the customer's choice. These locations are given as an indication.

Work executed to the complete work; Including supply, installation and connection, supply and installation of cables and all subjects.

No added value will be requested in the event of an omission in the description of an element, a service or a quality characteristic necessary for the proper functioning of the work.

PRICE PAID TO THE WHOLE.

#### ▪ **Cabling and Commissioning**

##### **-Price No. 15: Cabling and Commissioning**

The system will be cabled and connected following supplier instructions. Pre-installation testing will be carried out using equipment provided by the system supplier.

Includes:

- Complete network cabling and tubing
- Cabling tests
- Mounting and connection of all equipment
- All interfaces, accessories, connectors, and tools necessary for system functionality
- On-site personnel training
- Full technical documentation, including system architecture, technical manuals, connection diagrams, operating instructions, and other necessary documents

The work, including all supply, installation, and accessories, will be paid as a complete block. No added value will be requested in the event of an omission in the description of a component or accessory necessary for the proper functioning of the structure.

▪ **External Air Extraction Fan**

**Price No. 16:** External Air Extraction Fan – Remote Motor (Silent Operation)

This price rewards:

The supply and installation of a remote-mounted external extraction fan, intended to be connected to an existing domestic kitchen hood via a duct.

The fan shall be installed on the façade or rooftop, complete with weatherproof casing in corrosion-resistant coated steel or aluminum, compliant with a minimum of IPX4 (recommended IPX5).

The mounting on a stable, vibration-free surface using anti-vibration supports and appropriate wall brackets or roof curb, included in the delivery.

A motor asynchronous with integrated thermal protection, operating at 230 V – 50 Hz, and delivering an airflow rate between 600 m<sup>3</sup>/h, depending on hood capacity.

Noise attenuation measures ensuring ≤ 25 dB(A) inside the kitchen, including correct distancing and vibration isolation.

Electrical wiring from the distribution board or junction point, using cables of appropriate section, including:

- Power supply cable
- Grounding
- Local disconnect switch installed nearby

Connection to existing ductwork (Ø 125 or Ø 150 mm) with airtight jointing and clamps or flanges as needed.

Integration of a non-return backdraft shutter to prevent air ingress when idle.

All necessary sleeves, fasteners, grommets, terminals, ducts, wiring paths, and weatherproofing materials for full operational readiness.

Work paid for individually, including installation, mounting accessories, electrical and duct connections, commissioning, and all necessary components for proper operation.

No added value will be requested in the event of an omission in the description of a component or accessory necessary for the proper functioning of the system.

PRICE PAID PER UNIT.

- **Bill Of Quantities**

A detailed price breakdown follows the specifications, encompassing all work elements required for a fully operational and compliant electrical installation.

## **II. LOT: PLUMBING SYSTEMS**

- **Price No. 1 - PPR Pipes PN20**

The cold and hot water distribution network within the building (basement and technical shafts) will be constructed using polypropylene (PPR) pipes, mounted on MUPRO-type supports within a shaft or wall trench. Pipes will be of the PPR type, brand NIRON or equivalent, with a 20-bar rating.

Includes all necessary accessories (tees, connectors, etc.) for connecting various outlets.

Pricing includes fittings, pipe insulation, galvanized steel suspension clamps and supports, joints, and all necessary materials and installation work.

It is noted that the filling of any bleeding will be carried out by the Contractor according to the rules of the art.

Work paid by the linear metre for tubes and by the unit for shut-off valves including bending, cutting, fittings, clamps, supports, as well as all supplies and conditions of supplies and installation at price n°1:

- a) Ø40x6.7.
- b) Ø32x5.4.
- c) Ø25x4.2
- d) Ø20x3.4.
- e) Shut-off valves D32
- f) Shut-off valves D25
- g) Shut-off valves D20

- **Price No. 2 - PVC Pipes**

Sanitary device drains to vertical risers or collectors will be constructed using Class 1 PVC pipes joined by adhesive. Pipes will be of the brand FERROPLAST or equivalent, with a minimum thickness of 3.2 mm.

- Sink drains: Ø40 mm

Paid per linear meter, including bends, fittings, adhesive joints, siphons, cleanout plugs, and all necessary supplies and installation work.

Pipe Sizes:

- Ø33.6/40 mm
- Ø43.6/50 mm
- Ø68.6/75 mm

▪ **Price No. 3 - Double-Basin Sink**

Brand: ROCA, E-series, or equivalent.

- Includes a double-basin sink with a 1.80m x 0.60m stainless steel drainboard, minimum thickness 8/10 mm.
- Mounted on a masonry support (aggregate blocks) and equipped with:
  - Chrome-plated GROHE Eurostyle mixer tap with swivel spout and hand shower.
  - Chain-stopper siphon and grille.
  - Sealed against masonry using white cement and silicone mastic.
- Soundproofing: Coated with "BLACKSON" type bitumen.

Paid per unit, including sink, mixer tap, hand shower, chain stopper, siphon, joints, rosettes, installation, and connections.

▪ **Price No. 4 - 100L Electric Water Heater**

Supply, installation, connection and installation of an electric water heater assembly consisting of:

- A vertical electric water heater of the DEDIETRICH brand or equivalent.
- Equipped with GRK brand safety group or equivalent, air purge, tank insulation, etc.
- Shielded heating assembly with resistor on removable barrels, adjustable thermostat internally protected against corrosion by magnesium anode, white enamelled protective cover.
- NICOLL type PVC siphon funnel for connection to the DN 30/32 drain and connection of the drain to the first fall party or manhole.
- Supply and Connection EF and EC in polyethylene reticle diameter 16/20 from the manifold to the sanitary including fittings, polyethylene.
- EF and EC connection made of galvanised iron tube, including chrome-plated rosettes.
- 2 DN 25 isolation valves.
- Electrical connection and earthing.

Note:

- Labels specifying the choice of material must remain visible for the duration of the work.
- The sanitary appliances must be protected for the duration of the work, they will be cleaned at the end of the site by the Contractor of this lot without added value.
- Chrome rosettes included in the price will be placed at the outlet of each recessed pipe. - Glazed vat.
- The resistance will be plunged into a sheath
- Thermostat for regulation and safety.
- Food-grade interior coating.
- Electrical connection.

Paid as a complete unit, including mounting, fixing, hydraulic, and electrical connections.

▪ **Price No. 5 - Circular Galvanized Steel Duct Ø160**

The galvanized circular duct is made from a galvanized strip wound in a helix and stapled on itself. It comes in standard length of 3 ml, up to 1500 mm in diameter, and it is always female thus allowing the accessories to fit together.

The ducts and their fittings are made of Z275 galvanised steel sheet, whose tube and fitting thicknesses comply with the XP E 51-620 standard.

The ducts will be supported by means of galvanized steel support feet

The thicknesses used for the pipes and fittings comply with the "Icat International" standards. The part numbers of the most commonly used fittings (subject to availability) are delivered in stamped form.

Degreasing and protection are the same as rectangular ducts. Stapled spiral sheath made of galvanised steel, including sleeves, elbows, tees, taps, clamps, SIKLA brand brackets, sealing strips, including protection of the sheath on the terrace (special insulation), the conditions must be made of non-combustible materials.

- **Bill Of Quantities**

A detailed price breakdown follows the specifications, encompassing all work elements required for a fully operational and compliant electrical installation.